BUSINESS SUMMARY

Brain Tumor MRI Image Classification

AI-assisted triage and second-reader support for brain MRI - what was built, what it delivers and what must happen before clinical use

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87.5%	~97%	65%	50 ms
Test accuracy	Tumor-vs-normal detection*	Meningioma recall	Lighter model latency

*Approximate, derived from per-class precision and support; confirm with the exact confusion matrix.

The opportunity

Radiology teams face growing MRI volumes. Reviewing every scan in arrival order delays the cases that matter most, and distinguishing tumor types from normal scans is subject to inter-reader variability. An automated first look that ranks scans and offers a consistent second opinion can shorten turnaround and extend specialist-level support to under-resourced settings.

The solution

A deep-learning classifier trained on 2,443 labelled MRI slices that assigns each scan to glioma, meningioma, pituitary tumor or no tumor, and returns a confidence score. It is delivered as a web application that also shows a Grad-CAM heatmap, class probabilities and a stability check, so a clinician can judge the output instead of trusting a black box.

Headline results

Measure	Result	What it means for the business
Accuracy / macro-F1	87.5% / 0.864	Reliable enough to prioritise scans; not enough to replace a reader
ROC-AUC (one-vs-rest)	0.976	Strong ranking quality - the alert threshold can be tuned (e.g. a high-sensitivity triage mode) without retraining
Weakest class	Meningioma, recall 64.8%	About 1 in 3 meningiomas is missed - the main safety gap to close
Pituitary	Recall 100%, precision 81%	Nothing missed, but some false alarms add follow-up workload
Speed and size	112 ms (24 M params); lighter alternative 50 ms (2.9 M params) at 0.9 pp lower macro-F1	Runs on a standard CPU - no specialised hardware needed for a pilot

Where it creates value

Use case	How the tool helps
Early detection and patient triage	Flags scans likely to contain a tumor so specialists see them first, improving workflow and patient prioritisation
AI-assisted diagnosis	Gives radiologists a fast, consistent pre-read with an explanation heatmap, supporting turnaround time

Use case	How the tool helps
Second opinion in tele-medicine	Brings decision support to remote or under-resourced sites with few specialists
Research and clinical trials	Segments image archives by tumor type to speed cohort building

Risks and how they are managed

Risk	Status and mitigation
Inflated accuracy from data leakage	Found and fixed - the vendor split leaked 27% of test images; a group-aware re-split gives an honest estimate
Model reads image brightness or framing, not pathology	A six-descriptor baseline already scores ~0.80 macro-F1; Grad-CAM review and external validation required
Missed tumors (false 'no tumor')	About 9 of 327 tumor scans estimated missed; use as triage only, with radiologist review of every result
Narrow evidence base	One public dataset, 2-D slices, no patient IDs - cannot support clinical claims
Regulatory and liability	Research prototype, clearly labelled not a medical device; any clinical use needs regulatory approval

Readiness and recommendation

> **Verdict:** ready for a controlled, non-clinical pilot as a triage and second-reader *demonstration*. It is **not** ready for patient-facing or diagnostic use. A live app, reproducible pipeline and documented limitations are in place.

Recommended next steps

- **Validate externally** on multi-centre data with patient-level splits - the gating item for any real deployment.
- **Close the meningioma gap** with more data, class-specific thresholds and error review.
- **Add calibration and an 'uncertain - refer' band** so confidence becomes an actionable triage signal.
- **Compare tuned ResNet50V2 with tuned MobileNetV2** and choose the best accuracy-per-cost option for hosting.
- **Pilot with radiologists in the loop** and measure turnaround time and agreement before scaling.

Live demo: Streamlit app | Code: GitHub. Research prototype - not a medical device.